

# EBN111 Practical 2 Instructions

(To be completed during the practical session)

2017

You need to complete this during the practical session.

After doing this assignment and writing down all your answers, log on to ClickUP and complete the **Practical 2 Online Evaluation**.

**NB:**

- Use a full stop for denoting a decimal point. **DO NOT** use a comma.
- Use 2 decimal digits for all answers.
- Express your answers in the specified units.
- **DO NOT** supply units on ClickUp.
- The laboratory instruments are sensitive and expensive, use with care.

## **Circuit troubleshooting list:**

- *Sometimes resistor packages have incorrect labels! Measure your resistors (disconnected from the circuit) with the multimeter to make sure they are correct. You can also use the colour bands to quickly verify the resistor size.*
- *Check your breadboard circuit. If you have resistors with both legs plugged into the same node (short vertical slot), it is being **SHORTED** and does not form part of your circuit. Check the pre-prac slides for examples on this.*
- *Twisting resistor legs together does **NOT** form a reliable connection. Use your breadboard. - Press **Auto Scale** on the function generator to scale a waveform. If the waveform still does not look right, your wave generator/wiring is probably not setup correctly.*
- ***DO NOT** try to measure current in line with the multimeter. Instruction for determining current are clearly given in the questions.*
- *If the power supply **OVERLOAD** light goes on, you are shorting out the power supply and something will melt/set on fire. Immediately switch off and check your circuit connections. - Only use the black **COM** output on the power supply as your ground. The green output is **Earth** and should not be used.*

## **Introduction**

The aim of this practical is to introduce you to the basic equipment used when building electric circuits. You will build a few simple electric circuits using simple passive electronic components in order to test a number of basic electronic concepts.

It is important that you understand the theory before building and testing your circuits.

Make sure that you have read through the **Practical 2 Laboratory Guide** before starting with this practical.

You will need to use the components in the box that you have received.

## EBN122 Practical 2

### Question 1

[8]

#### Objective

This question introduces the power supply and the multimeter. The goal here is to realize that a measured value and the theoretical value is not always 100% the same.

#### Instructions

1. Before you switch on the power supply, make sure all the voltage knobs are turned down to the minimum. The voltage is decreased by rotating the knob counter clockwise and increased by rotating the knob clockwise.
2. Connect the power supply directly to the multimeter as shown in Figure 1a and press button **A** as indicated in Figure 1a on the left. Remember that you are measuring DC voltage, so make sure the multimeter is set to measure this, measured by pressing button **B** indicated in Figure 1a. Use to **Banana plug cables** end to end.
3. After setting up the equipment, turn on the power supply.
4. Complete Table 1. (*Note: the measured voltages will not correspond EXACTLY with the output voltages, the point is to get close!*)

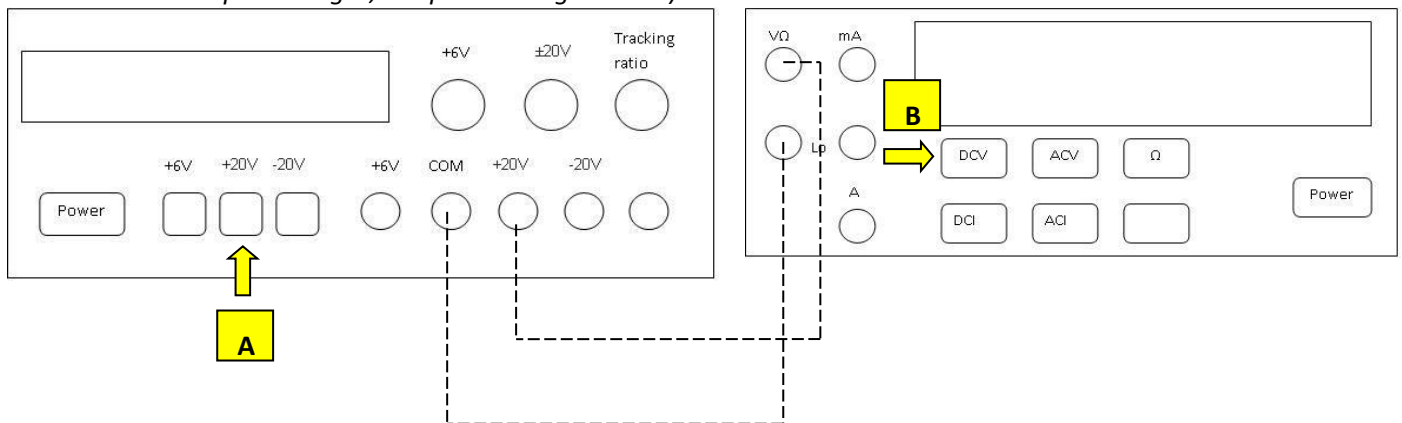


Figure 1a. Connecting the multimeter to the power supply.

| Voltage supplied [V]<br>(Set the power supply to these values) | Voltage measured [V]<br>(Read off from the multimeter) |
|----------------------------------------------------------------|--------------------------------------------------------|
| 3.3                                                            |                                                        |
| 6.5                                                            |                                                        |
| 9                                                              |                                                        |
| 18                                                             |                                                        |

Table 1. Voltage supplied and voltage measured.

## Question 2

[6]

### Objective

This question involves measuring resistance using the multimeter.

### Instructions

Find a **560  $\Omega$** , **12 k $\Omega$**  and **270 k $\Omega$**  resistor and measure the resistances using the multimeter.

(Note: A **1k2  $\Omega$**  resistor refers to **1200  $\Omega$** . Be careful not to mix up **12  $\Omega$** , **1k2  $\Omega$**  and **12 k $\Omega$** )

1. Connect the resistor to as shown in Figure 1b below.
2. After setting up the equipment, turn on the power supply.
3. Press button **C** indicated in Figure 1b to measure the resistance.
4. Complete Table 1.

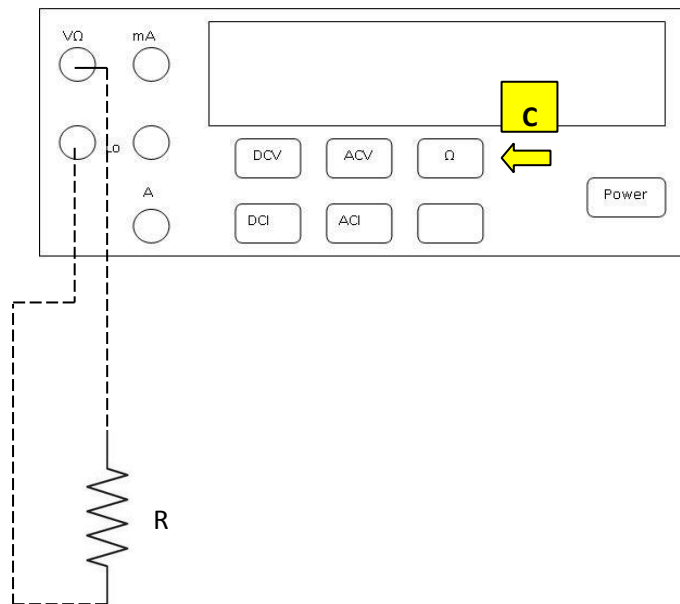


Figure 1b. Connecting the multimeter to the resistor.

| Defined resistance<br>(From the colour code on the resistor) |            | Measured resistance (Read off multimeter) |            |
|--------------------------------------------------------------|------------|-------------------------------------------|------------|
| 560                                                          | $\Omega$   |                                           | $\Omega$   |
| 12                                                           | k $\Omega$ |                                           | k $\Omega$ |
| 270                                                          | k $\Omega$ |                                           | k $\Omega$ |

Table 2. Defined resistance and measured resistance.

### Question 3

[14]

#### Objective

In this question you will monitor a signal generated by the function generator using the oscilloscope.

#### Instructions

Connect the equipment as shown in Figure 2 and follow the steps below.

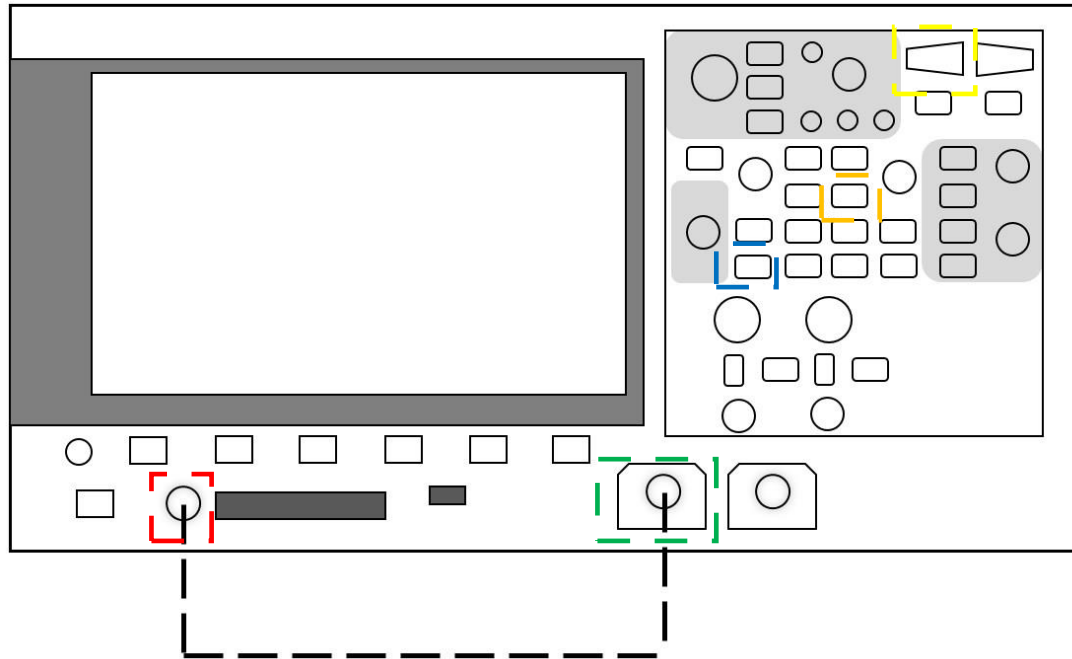


Figure 2. Connecting the function generator to the oscilloscope.

1. Connect the **Gen Out** (highlighted in red) to the **Channel 1** (highlighted in green) of the oscilloscope. Use the **two sets** of BNC cables and connect the crocodile clip ends black to black and red to red.
2. Switch on the oscilloscope.
3. Push the **Wave Gen** button on the oscilloscope (highlighted in blue).
4. Select the **Waveform** to be a sine wave with frequency **20 kHz**, **1.5Vpp** Amplitude and **0 V** offset.
5. Now scale your graph until you clearly see the sine wave on the screen. Use the **Horizontal** and **Vertical** knobs to do this (push to zero). You can also use the **Auto scale** button to do this for you.
6. Push the **Meas** button on your scope (highlighted in orange).
7. Add the following measurements: Frequency, period and rise-time. Do this by selecting the **Type** and then adding the measurement
8. Press the **Run/Stop** button (highlighted in yellow).
9. Complete Table 3.

| Measured frequency [kHz] | Measured period [ $\mu$ s] | Measured rise-time [ $\mu$ s] |
|--------------------------|----------------------------|-------------------------------|
|                          |                            |                               |

Table 3. Values measured using the oscilloscope.

Question 4

[6]

**Objective**

In this question you will test the principle of voltage division. Figure 3 illustrates the concept. You will solve the circuit theoretically and then measure the value by building the actual circuit.

**Instructions**

For  $V_G = 10\text{ V}$ ,  $R_1 = 56\text{ k}\Omega$  and  $R_2 = 120\text{ k}\Omega$ . Calculate  $V_2$  theoretically by hand (using Ohm’s law and voltage division). Write your answer in Table 4 below.

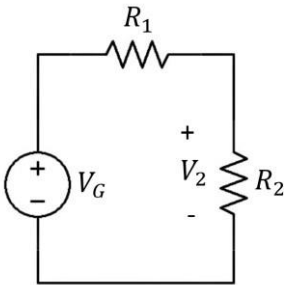


Figure 3. Circuit diagram for Question 4.

Now **build** the circuit on your **breadboard** and measure the voltage  $V_2$ , refer to Figure 4. Set your power supply to the voltage of  $V_G$  (10 V). Complete Table 4.

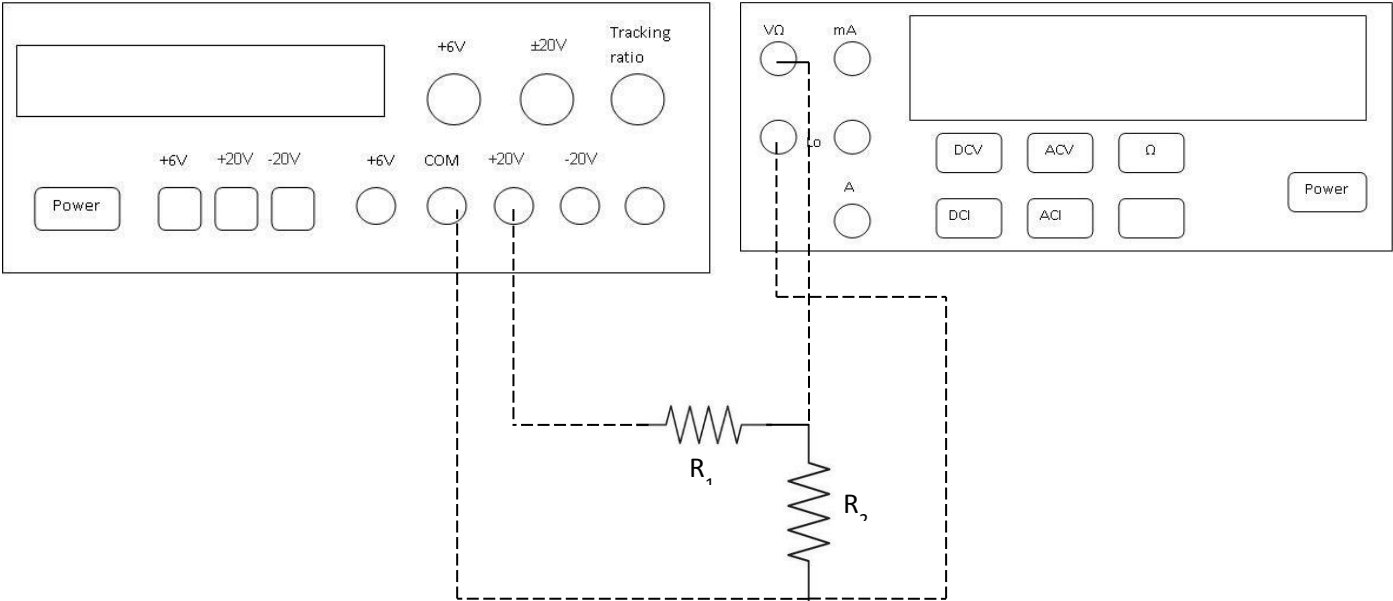


Figure 4. Wiring diagram for Figure 3.

| Theoretical value for $V_2$ [V] | Measured value for $V_2$ [V] |
|---------------------------------|------------------------------|
|                                 |                              |

Table 4. Theoretical and measured values for the voltage  $V_2$ .

## Question 5

[12]

### Objective

In this question you will test the concept of current division. Figure 5 illustrates the concept. You will solve the circuit theoretically and then build the actual circuit. You will then measure the voltage and resistance and use Ohm's Law to calculate the current values.

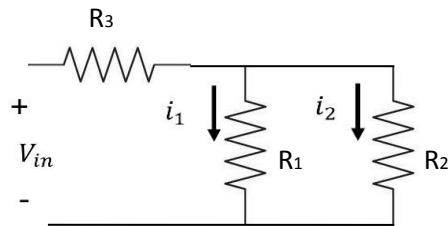


Figure 5. Circuit diagram for Question 5.

### Instructions

For  $R_1 = 12 \text{ k}\Omega$ ,  $R_2 = 56 \text{ k}\Omega$  and  $R_3 = 5 \text{ k}\Omega$  and  $V_{in} = 6 \text{ V}$ , determine the theoretical values of  $i_1$  and  $i_2$  using Ohm's law and current division. (Note: You can also use the +6V button and the +6V voltage output to fine tune your supply)

| Theoretical value for $i_1$ [ $\mu\text{A}$ ] | Theoretical value for $i_2$ [ $\mu\text{A}$ ] |
|-----------------------------------------------|-----------------------------------------------|
|                                               |                                               |

Table 5. Theoretical values for  $i_1$  and  $i_2$ .

Now build the circuit and measure the actual voltages  $V_1$  and  $V_2$ . Figure 6 shows the circuit you must build to measure  $V_2$ . Complete Table 6. Use Ohm's law to calculate  $i_1$  and  $i_2$ .

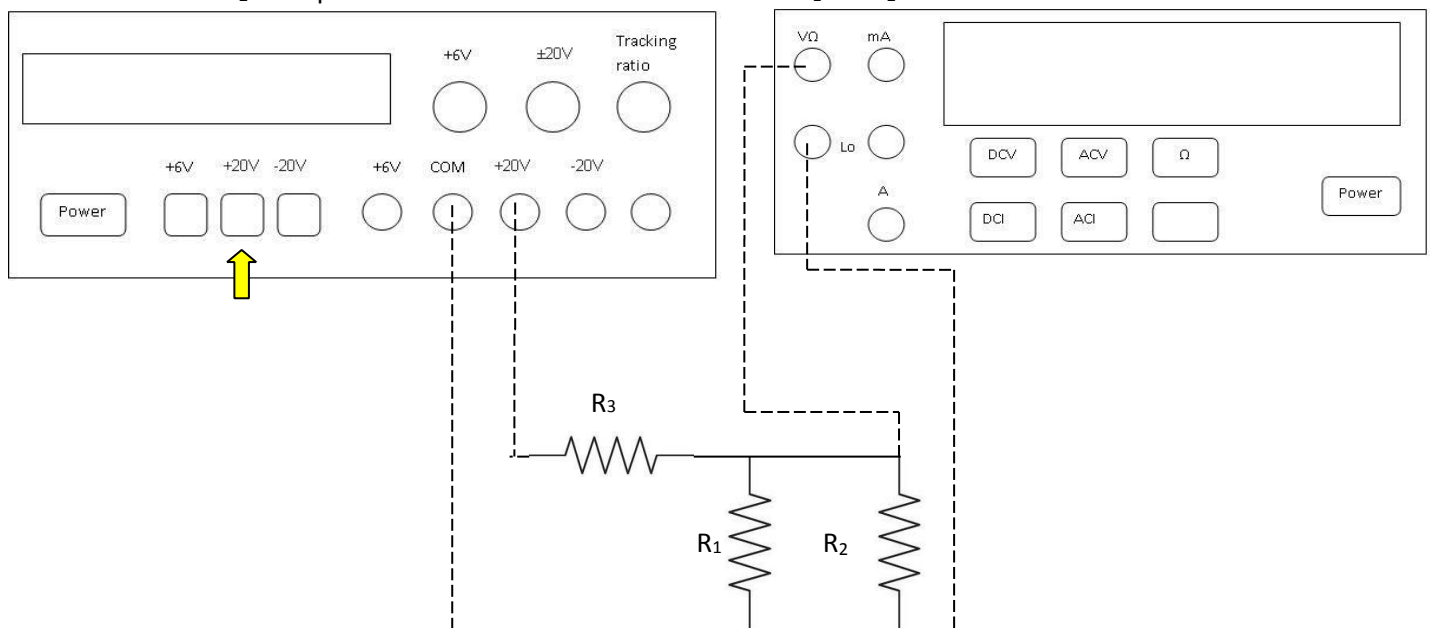


Figure 6. Wiring diagram for Figure 5 where  $i_2$  is measured

| Measured value for $i_1$ [ $\mu\text{A}$ ] | Measured value for $i_2$ [ $\mu\text{A}$ ] |
|--------------------------------------------|--------------------------------------------|
|                                            |                                            |

Table 6. Measured values for  $i_1$  and  $i_2$ .

Question 6

[16]

Instructions

Build the circuit and measure the node voltage  $V_1$ , voltage  $V_o$  shown in Figure 7 and determine the current  $i_x$  by **measuring the resistance of R1 as well as the voltage across R1**. Use these values to **determine  $i_x$  with Ohm's Law**. Complete Table 7.  $V_{in} = 12V$

R1 = 12kΩ

R3 = 27kΩ

R2 = 2k7Ω

R4 = R5 = 56 kΩ

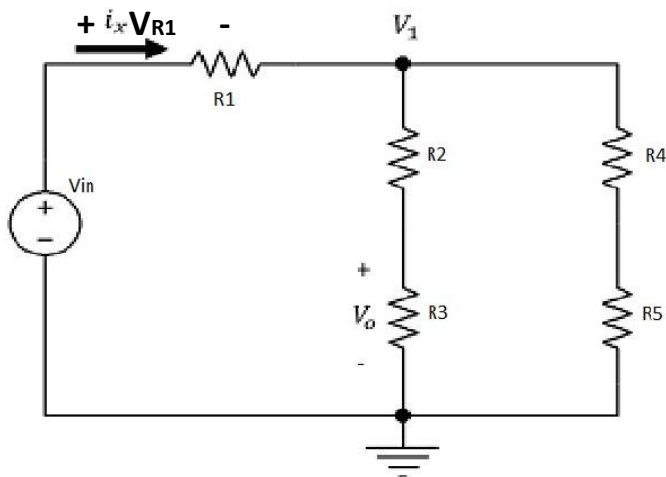


Figure 7. Circuit diagram for Question 6.

| $i_x$ [μA] | $V_1$ [V] | $V_o$ [V] |
|------------|-----------|-----------|
|            |           |           |

Table 7. Measured values for Question 6.

Question 7

[4]

Instructions

Build the circuit in Figure 8, with resistance R as defined in your Online Evaluation. Measure  $R_{eq}$  using the multimeter. Complete Table 8.

**Important**  
The value of R is randomly chosen on ClickUp. Open the Practical 2 Online Evaluation to see your value

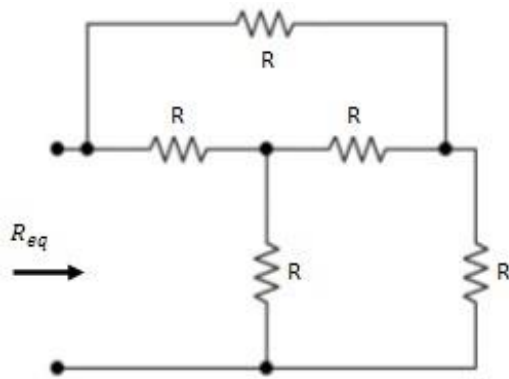


Figure 8. Circuit diagram for Question 7.

| $R_{eq} [k\Omega]$ |
|--------------------|
|                    |

Table 8. Equivalent resistance of the circuit in Figure 8.

Question 8

[14]

**Instructions**  
 Build the circuit in Figure 9 and measure the values of  $i_x$  and  $V_o$ . Remember that in order to measure  $i_x$ , you must first **measure the voltage over R6 and the resistance of R6 and then use Ohm's law** to calculate the current. Complete Table 9.  
 $V_{in} = 15V$   
 $R1 = 12\text{ k}\Omega$   
 $R2 = 27\text{ k}\Omega$   
 $R3 = 120\text{ k}\Omega$   
 $R4 = R5 = 56\text{ k}\Omega$   
 $R6 = 5\text{ k}\Omega$

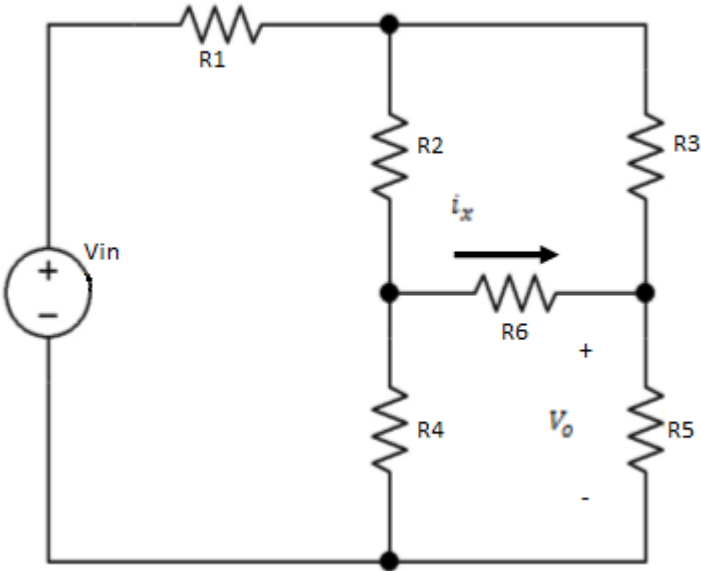


Figure 9. Circuit diagram for Question 8.

| $i_x [\mu A]$ | $V_o [V]$ |
|---------------|-----------|
|               |           |

Table 9. Measured values for the circuit in Figure 9.

Now enter all your answers on ClickUp in **Practical 2 Online Evaluation**.